

HA NOI, Viet Nam

WMO: 488200

Lat: 21.0205N

Lon: 105.8070E

Elev: 12

StdP: 101.18

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.0	(c) 11.0	(d) 2.0	(e) 4.4	(f) 14.6	(g) 4.1	(h) 5.1	(i) 14.6	(j) 7.6	(k) 18.0	(l) 7.0	(m) 18.1	(n) 2.7	(o) 50	(p) 0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 7.6	(c) 36.2	(d) 27.4	(e) 35.1	(f) 27.4	(g) 34.2	(h) 27.3	(i) 29.4	(j) 32.6	(k) 28.9	(l) 32.2	(m) 28.4	(n) 31.7	(o) 2.7	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.9	25.5	30.8	28.1	24.3	30.4	27.8	23.9	30.2	97.1	32.3	94.6	32.0	92.2	31.8	35.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.0	(b) 6.1	(c) 5.4	(e) 8.0	(f) 38.6	(g) 1.4	(h) 1.3	(i) 6.9	(j) 39.6	(k) 6.1	(l) 40.4	(m) 5.3	(n) 41.1	(o) 4.2	(p) 42.1		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			8.0	30.5	1.3	1.3	5.3	31.5	4.5	32.3	3.7	33.0	2.8	34.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	17.1	18.7	21.0	25.0	28.1	29.9	29.7	29.1	28.2	26.1	22.7	18.7
	DBStd	5.29	3.65	4.11	3.31	2.89	2.40	1.85	1.73	1.62	1.69	2.24	2.93	3.15
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	163	69	44	13	1	0	0	0	0	0	0	3	34
	CDD10.0	5309	221	243	340	449	562	597	611	593	547	499	379	268
	CDD18.3	2430	32	53	94	200	304	347	353	334	297	241	132	44
	CDH23.3	24920	70	178	354	1496	3386	4514	4519	4035	3311	2148	764	146
(14) Wind	WSAvg	2.5	2.5	2.8	2.7	2.9	2.9	2.6	2.6	2.3	2.2	2.2	2.3	2.4
	PrecAvg	1652	20	28	44	99	186	248	265	302	241	151	59	14
	PrecMax	2246	76	92	260	268	551	523	492	665	562	469	259	54
	PrecMin	1033	1	4	6	10	29	40	102	39	42	10	0	0
	PrecStd	289	19	22	43	55	101	109	104	134	106	121	58	16
	DB	27.1	29.2	30.2	34.2	37.7	38.1	37.2	36.1	35.1	33.2	31.7	28.8	
	MCWB	22.1	23.3	24.6	27.3	27.4	27.2	27.7	27.4	26.4	24.6	24.4	22.5	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	25.1	27.1	28.1	31.9	35.2	36.4	35.8	34.9	33.9	32.2	30.0	26.8	
	2% MCWB	20.9	22.3	23.8	26.2	27.5	27.4	27.6	27.4	26.6	24.5	23.4	21.6	
	5% DB	23.8	25.3	26.4	30.2	33.8	35.1	34.8	33.9	33.0	31.2	28.8	24.9	
	5% MCWB	20.1	21.7	23.1	25.7	27.4	27.4	27.5	27.5	26.3	24.2	22.7	20.7	
	10% DB	22.1	24.0	25.1	29.0	32.2	34.0	33.7	32.9	32.0	30.2	27.1	23.3	
	10% MCWB	19.3	21.4	22.7	25.1	26.7	27.5	27.4	27.4	25.8	24.1	21.9	19.6	
	WB	23.1	24.2	25.7	28.2	29.6	29.9	30.0	30.0	29.2	27.3	26.0	24.1	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	25.4	26.8	27.8	31.1	33.1	33.5	33.3	32.5	31.8	30.0	28.9	26.2	
	2% WB	21.9	23.2	24.6	27.1	28.7	29.2	29.1	29.1	28.2	26.5	25.1	23.0	
	2% MCDB	24.1	25.7	26.8	30.0	32.4	32.6	32.6	32.0	30.9	29.2	27.9	25.2	
	5% WB	21.0	22.5	23.8	26.3	28.1	28.7	28.6	28.5	27.7	26.0	24.4	21.9	
	5% MCDB	22.9	24.8	25.8	29.0	31.7	32.1	32.0	31.5	30.4	28.7	27.0	24.1	
	10% WB	20.1	21.8	23.2	25.7	27.5	28.2	28.1	27.9	27.1	25.5	23.6	20.7	
	10% MCDB	21.8	23.8	24.9	28.3	31.0	31.6	31.4	31.0	29.9	28.2	26.1	22.9	
(35) Mean Daily Temperature Range	MDBR	5.5	5.2	5.0	6.0	7.3	7.6	7.0	6.7	6.8	7.1	7.1	6.5	
	5% DB	7.6	6.9	6.6	7.8	9.2	9.2	8.6	8.3	8.2	8.3	8.5	8.0	
	MCDBR	3.8	3.2	3.4	3.8	3.9	3.5	3.6	3.5	3.2	3.0	3.5	3.7	
	5% WB	6.5	6.2	5.8	6.8	8.1	8.0	7.6	7.3	7.1	7.3	7.4	6.6	
	MCWBR	4.0	3.3	3.5	3.9	4.1	3.7	3.8	3.8	3.5	3.3	3.8	4.1	
(40) Clear-Sky Solar Irradiance	taub	0.610	0.642	0.816	0.776	0.566	0.521	0.493	0.553	0.607	0.712	0.614	0.641	
	taud	1.718	1.695	1.426	1.529	2.015	2.157	2.230	2.037	1.862	1.613	1.760	1.680	
	Ebn at Noon	661	674	582	618	758	786	808	765	716	613	653	615	
	Edh at Noon	218	235	320	292	178	153	142	173	204	252	207	220	
(44) All-Sky Solar Radiation	RadAvg	2.27	2.69	2.98	3.82	4.80	5.00	4.85	4.62	4.38	3.77	3.29	2.55	
	RadStd	0.49	0.45	0.34	0.40	0.46	0.41	0.42	0.35	0.30	0.29	0.43	0.40	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-1.20	+0.77	N/A	N/A	N/A	N/A	N/A	+110
(47) Regional (6 neighbors)	N/A	N/A	N/A	+0.58	+0.18	N/A	N/A	N/A	N/A	+80

Nomenclature: See separate page